



**Laboratorium  
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*Institut Teknologi Sepuluh Nopember***

# **Laporan Akhir Praktikum Jaringan Komputer**

## **Vlan Trunk OSPF MultiVendor**

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2026

# 1 Langkah-Langkah Percobaan

Berikut adalah langkah-langkah yang dilakukan pada praktikum Jaringan KOmputer P5, yang mana terbagi menjadi beberapa bagian praktikum.

## 1.1 Praktikum 1: Konfigurasi Cisco Switch

1. Masuk ke mode konfigurasi dan menambahkan VLAN 10, 20, 30 dan 40.

```
Switch>enable
Switch#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name Finance
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name HR
Switch(config-vlan)#exit
Switch(config)#vlan 30
Switch(config-vlan)#name IT
Switch(config-vlan)#exit
Switch(config)#vlan 40
Switch(config-vlan)#name Marketing
Switch(config-vlan)#exit
Switch(config)#
```

**Gambar 1:** Menambahkan VLAN 10, 20, 30, dan 40 pada switch Cisco

2. Memastikan bahwa VLAN sudah dibuat dan statusnya aktif

```
Switch#wr
Building configuration...
Compressed configuration from 2905 bytes to 1378 bytes[OK]
Switch#
*Jun  5 02:20:46.069: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk. Please wait...
*Jun  5 02:20:46.719: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.
Switch#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Gi0/0, Gi0/1, Gi0/2, Gi0/3
10   Finance                 active    Gi1/0, Gi1/1, Gi1/2, Gi1/3
20   HR                      active
30   IT                       active
40   Marketing                active
1002 fddi-default          act/unsup
1003 token-ring-default    act/unsup
1004 fddinet-default        act/unsup
1005 trnet-default          act/unsup
Switch#
```

**Gambar 2:** Verifikasi VLAN aktif dengan show vlan brief

3. Mengkonfigurasi access port untuk setiap VLAN

```
Switch(config)#
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int gi0/2
Switch(config-if)#description PC-FINANCE-VLAN 10
Switch(config-if)#description PC-FINANCE-VLAN10
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#
```

**Gambar 3:** Konfigurasi access port VLAN 10 pada interface Gi0/2

```
Switch(config)#int gi0/1
Switch(config-if)#description PC-HR-VLAN20
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#
```

**Gambar 4:** Konfigurasi access port VLAN 20 pada interface Gi0/1

```
Switch(config)#int gi1/1
Switch(config-if)#description PC-IT-VLAN30
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#
```

**Gambar 5:** Konfigurasi access port VLAN 30 pada interface Gi1/1

```
Switch(config-if)#int gi0/3
Switch(config-if)#description PC-Marketing-VLAN40
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 40
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#
```

**Gambar 6:** Konfigurasi access port VLAN 40 pada interface Gi0/3

#### 4. Mengkonfigurasi trunk ke cisco router dan memverifikasinya.

```
Switch(config)#int gi0/0
Switch(config-if)#switchport trunk encapsulation dot1q
Switch(config-if)#description TRUNK-T0-CISCO-ROUTER
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk allowed vlan 10,20,30,40
Switch(config-if)#no shutdown
Switch(config-if)#exit
```

**Gambar 7:** Konfigurasi trunk Gi0/0 ke router Cisco

```
*Jun  5 02:25:20.832: %SYS-5-CONFIG_I: Configured from console by console
Switch#wr
Building configuration...
Compressed configuration from 3364 bytes to 1645 bytes[OK]
Switch#
*Jun  5 02:25:24.744: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk. Please wait...
*Jun  5 02:25:25.404: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.
Switch#show interface trunk

Port      Mode      Encapsulation  Status        Native vlan
Gi0/0     on        802.1q         trunking      1

Port      Vlans allowed on trunk
Gi0/0     10,20,30,40

Port      Vlans allowed and active in management domain
Gi0/0     10,20,30,40

Port      Vlans in spanning tree forwarding state and not pruned
Gi0/0     10,20,30,40
Switch#
```

**Gambar 8:** Verifikasi trunk dengan show interface trunk dan write memory

#### 5. write memory

(Hasil write memory sudah ditampilkan bersama Gambar 8)

## 1.2 Praktikum 2: Konfigurasi Cisco Router

#### 1. Konfigurasi interface trunk ke switch

```
Router#enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int gig0/2
Router(config-if)#description TRUNK-T0-CISCO-SWITCH
Router(config-if)#no ip address
Router(config-if)#no shutdown
Router(config-if)#exit
```

**Gambar 9:** Konfigurasi interface trunk router Cisco menuju switch

#### 2. Konfigurasi subinterface VLAN 10

```
Router(config)#int gig0/2.10
Router(config-subif)#description GATEWAY-VLAN10-FINANCE
Router(config-subif)#encapsulation dot10 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
```

**Gambar 10:** Konfigurasi subinterface Gi0/2.10 untuk VLAN 10

### 3. Konfigurasi subinterface VLAN 20

```
Router(config)# int gig0/2.20
Router(config-subif)#description GATEWAY-VLAN20-HR
Router(config-subif)#encapsulation dot10 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
```

**Gambar 11:** Konfigurasi subinterface Gi0/2.20 untuk VLAN 20

### 4. Konfigurasi subinterface VLAN 30

```
Router(config)#int gig0/2.30
Router(config-subif)#description GATEWAY-VLAN30-IT
Router(config-subif)#encapsulation dot10 30
Router(config-subif)#ip address 192.168.30.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
```

**Gambar 12:** Konfigurasi subinterface Gi0/2.30 untuk VLAN 30

### 5. Konfigurasi subinterface VLAN 40

```
Router(config)#int gig0/2.40
Router(config-subif)#description GATEWAY-VLAN40-MARKETING
Router(config-subif)#encapsulation dot10 40
Router(config-subif)#ip address 192.168.40.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
```

**Gambar 13:** Konfigurasi subinterface Gi0/2.40 untuk VLAN 40

### 6. Mengkonfigurasi DHCP-Server untuk VLAN 10

```
Router(config)#ip dhcp pool VLAN10-FINANCE
Router(dhcp-config)#exit
Router(config)#ip dhcp excluded-address 192.168.10.1 192.168.10.10
Router(dhcp-config)#network 192.168.10.0
Router(dhcp-config)#network 192.168.10.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.10.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
```

**Gambar 14:** Konfigurasi DHCP pool untuk VLAN 10 (Finance)

### 7. Mengkonfigurasi DHCP-Server untuk VLAN 20

```
Router(config)#ip dhcp excluded-address 192.168.20.1 192.168.20.10
Router(dhcp-config)#ip dhcp pool VLAN20-HR
Router(dhcp-config)#network 192.168.20.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.20.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
```

**Gambar 15:** Konfigurasi DHCP pool untuk VLAN 20 (HR)

### 8. Mengkonfigurasi IP static VPCS pada VLAN 30, 40 dan 10. 20

*(Konfigurasi IP static/DHCP pada VPCS ditampilkan pada Gambar 19, 21, dan langkah ping berikutnya)*

### 9. Memverifikasi Cisco router

**Gambar 16:** Verifikasi router dengan show ip interface brief

**Gambar 17:** Mengecek DHCP pool dengan show ip dhcp pool

**Gambar 18:** Mengecek client DHCP dengan show ip dhcp binding

**Gambar 19:** PC VLAN 10 berhasil mendapatkan IP secara DHCP

**Gambar 20:** Ping dari PC VLAN 10 ke gateway 192.168.10.1

```

VPCS> ip dhcp
DDORA IP 192.168.20.11/24 GW 192.168.20.1
VPCS> show ip
NAME          : VPCS[1]
IP/MASK       : 192.168.20.11/24
GATEWAY       : 192.168.20.1
DNS           : 8.8.8.8
DHCP SERVER   : 192.168.20.1
DHCP LEASE    : 86163, 86400/43200/75600
MAC           : 00:50:79:66:68:40
LPORT        : 20000
RHOST:PORT    : 127.0.0.1:30000
RTU           : 1500
VPCS>

```

**Gambar 21:** PC VLAN 20 berhasil mendapatkan IP secara DHCP

15. Melakukan ping gateway 192.168.20.1 dari PC VLAN 20

```

VPCS> ping 192.168.20.1
84 bytes from 192.168.20.1 icmp_seq=1 ttl=255 time=1.533 ms
84 bytes from 192.168.20.1 icmp_seq=2 ttl=255 time=4.361 ms
84 bytes from 192.168.20.1 icmp_seq=3 ttl=255 time=2.658 ms
84 bytes from 192.168.20.1 icmp_seq=4 ttl=255 time=2.375 ms

```

**Gambar 22:** Ping dari PC VLAN 20 ke gateway 192.168.20.1

16. Uji ping dari PC VLAN 30 ke gateway 192.168.30.1

```

VPCS> ping 192.168.30.1
84 bytes from 192.168.30.1 icmp_seq=1 ttl=255 time=2.074 ms
84 bytes from 192.168.30.1 icmp_seq=2 ttl=255 time=3.526 ms
84 bytes from 192.168.30.1 icmp_seq=3 ttl=255 time=3.398 ms
84 bytes from 192.168.30.1 icmp_seq=4 ttl=255 time=4.063 ms

```

**Gambar 23:** Ping dari PC VLAN 30 ke gateway 192.168.30.1

17. Uji ping dari PC VLAN 40 ke gateway 192.168.40.1

```

VPCS> ping 192.168.40.1
84 bytes from 192.168.40.1 icmp_seq=1 ttl=255 time=2.392 ms
84 bytes from 192.168.40.1 icmp_seq=2 ttl=255 time=1.765 ms
84 bytes from 192.168.40.1 icmp_seq=3 ttl=255 time=2.671 ms
84 bytes from 192.168.40.1 icmp_seq=4 ttl=255 time=1.861 ms
84 bytes from 192.168.40.1 icmp_seq=5 ttl=255 time=2.528 ms

```

**Gambar 24:** Ping dari PC VLAN 40 ke gateway 192.168.40.1

18. Uji ping antar VLAN dari PC VLAN 10

```

VPCS> ping 192.168.20.11
192.168.20.11 icmp_seq=1 ttl=64 time=0.001 ms
192.168.20.11 icmp_seq=2 ttl=64 time=0.001 ms
192.168.20.11 icmp_seq=3 ttl=64 time=0.001 ms
192.168.20.11 icmp_seq=4 ttl=64 time=0.001 ms
192.168.20.11 icmp_seq=5 ttl=64 time=0.001 ms

VPCS> ping 192.168.30.10
84 bytes from 192.168.30.10 icmp_seq=1 ttl=63 time=7.146 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=63 time=3.105 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=63 time=2.675 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=63 time=3.266 ms
84 bytes from 192.168.30.10 icmp_seq=5 ttl=63 time=2.611 ms

VPCS> ping 192.168.40.10
84 bytes from 192.168.40.10 icmp_seq=1 ttl=63 time=5.050 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=63 time=2.260 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=63 time=2.488 ms
84 bytes from 192.168.40.10 icmp_seq=4 ttl=63 time=2.730 ms

```

**Gambar 25:** Ping antar VLAN dari PC VLAN 10 ke VLAN 20, 30, dan 40

19. Cek IP-DHCP binding pada cisco router

```

Router#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration        Type
                Hardware address/
                User name
192.168.10.11   0100.5079.6668.3f  Jun 06 2026 02:41 AM    Automatic
192.168.20.11   0100.5079.6668.40  Jun 06 2026 02:39 AM    Automatic
Router#

```

**Gambar 26:** Cek IP-DHCP binding pada router Cisco

### 1.3 Praktikum 3: Konfigurasi Huawei Router dan IP Address Antarrouter

#### 1. Mengkonfigurasi LAN Huawei

```
Enter system view, return user view with return command.
[~HUAWEI]
[~HUAWEI]interface Ethernet1/0/2
[~HUAWEI-Ethernet1/0/2] description LAN-HUAWEI
[*HUAWEI-Ethernet1/0/2] ip address 192.168.50.1 255.255.255.0
[*HUAWEI-Ethernet1/0/2] undo shutdown
[*HUAWEI-Ethernet1/0/2] quit
[*HUAWEI]
[*HUAWEI]commit
[~HUAWEI]
```

**Gambar 27:** Konfigurasi LAN Huawei pada interface Ethernet1/0/2

#### 2. Konfigurasi IP VPC LAN Huawei lalu memverifikasinya

```
VPCS> ip 192.168.50.10/24 192.168.50.1
Checking for duplicate address...
PC1 : 192.168.50.10 255.255.255.0 gateway 192.168.50.1

VPCS> show ip
NAME      : VPCS[1]
IP/MASK   : 192.168.50.10/24
GATEWAY   : 192.168.50.1
DNS       :
MAC       : 00:50:79:66:68:3e
LPORT    : 20000
RHOST:PORT : 127.0.0.1:30000
ttl       : 1500
```

**Gambar 28:** Konfigurasi IP VPC LAN Huawei dan verifikasi show ip

#### 3. Testing ping gateway Huawei

```
VPCS> ping 192.168.50.1
84 bytes from 192.168.50.1 icmp_seq=1 ttl=255 time=7.575 ms
84 bytes from 192.168.50.1 icmp_seq=2 ttl=255 time=2.284 ms
84 bytes from 192.168.50.1 icmp_seq=3 ttl=255 time=2.423 ms
^C
VPCS>
```

**Gambar 29:** Testing ping dari PC LAN Huawei ke gateway 192.168.50.1

#### 4. Konfigurasi IP link Huawei dengan Cisco

```
[~HUAWEI]int Ethernet1/0/0
[~HUAWEI-Ethernet1/0/0]description LINK1-T0-CISCO
[*HUAWEI-Ethernet1/0/0]ip address 10.10.10.2 255.255.255.252
[*HUAWEI-Ethernet1/0/0]undo shutdown
[*HUAWEI-Ethernet1/0/0]quit
[*HUAWEI]commit
[~HUAWEI]
```

**Gambar 30:** Konfigurasi IP link Huawei ke Cisco (Ethernet1/0/0, link 1)

#### 5. Konfigurasi IP link Huawei 2 dengan Cisco

```
Error: Unrecognized command found at '^' position.
[~HUAWEI]int Ethernet1/0/0
[~HUAWEI-Ethernet1/0/0]description LINK1-T0-CISCO
[*HUAWEI-Ethernet1/0/0]ip address 10.10.10.2 255.255.255.252
[*HUAWEI-Ethernet1/0/0]undo shutdown
[*HUAWEI-Ethernet1/0/0]quit
[*HUAWEI]commit
[~HUAWEI]int Ethernet1/0/4
[~HUAWEI-Ethernet1/0/4]description LINK2-T0-CISCO
[*HUAWEI-Ethernet1/0/4]ip address 10.10.11.2 255.255.255.252
[*HUAWEI-Ethernet1/0/4]undo shutdown
[*HUAWEI-Ethernet1/0/4]quit
[*HUAWEI]commit
[~HUAWEI]
```

**Gambar 31:** Konfigurasi IP link Huawei ke Cisco (Ethernet1/0/4, link 2)

#### 6. Konfigurasi IP link Huawei dengan Mikrotik

```

[~HUAWEI]int Ethernet1/0/3
[~HUAWEI-Ethernet1/0/3]description LINK-TO-MIKROTIK
[~HUAWEI-Ethernet1/0/3]ip address 10.10.20.1 255.255.255.252
[~HUAWEI-Ethernet1/0/3]undo shutdown
[~HUAWEI-Ethernet1/0/3]quit
[~HUAWEI]commit
[~HUAWEI]

```

**Gambar 32:** Konfigurasi IP link Huawei ke MikroTik (Ethernet1/0/3)

```

[admin@MikroTik] > /ip address add address= 10.10.20.2/30 interface=1 comment=T0-HUAWEI
[admin@MikroTik] >

```

**Gambar 33:** Konfigurasi IP link MikroTik ke Huawei (ether1)

## 7. Konfigurasi IP link Cisco dengan Huawei

```

Router#enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int gi0/3
Router(config-if)#description LINK1-TO-HUAWEI
Router(config-if)#ip address 10.10.10.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
*Jun 5 02:49:16.640: %LINK-3-UPDOWN: Interface GigabitEthernet0/3, changed state to up
*Jun 5 02:49:17.640: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/3, changed state to up
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
*Jun 5 02:49:26.471: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk. Please wait...
*Jun 5 02:49:27.146: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.

```

**Gambar 34:** Konfigurasi IP link Cisco ke Huawei (Gi0/3, link 1)

```

Router(config)#int gi0/0
Router(config-if)#description LINK2-TO-HUAWEI
Router(config-if)#ip address 10.10.11.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
*Jun 5 02:51:23.919: %LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to up
*Jun 5 02:51:24.920: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
*Jun 5 02:51:28.999: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk. Please wait...
*Jun 5 02:51:29.694: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.

```

**Gambar 35:** Konfigurasi IP link Cisco ke Huawei (Gi0/0, link 2)

## 8. Memverifikasi interface

```

Router#show ip interface brief
Interface          IP-Address      OK? Method Status  Protocol
GigabitEthernet0/0 10.10.11.1      YES manual up      up
GigabitEthernet0/1 10.10.30.1      YES manual up      up
GigabitEthernet0/2 unassigned      YES unset  up      up
GigabitEthernet0/2.10 192.168.10.1   YES manual up      up
GigabitEthernet0/2.20 192.168.20.1   YES manual up      up
GigabitEthernet0/2.30 192.168.30.1   YES manual up      up
GigabitEthernet0/2.40 192.168.40.1   YES manual up      up
GigabitEthernet0/3 10.10.10.1      YES manual up      up
Router#

```

**Gambar 36:** Verifikasi interface router Cisco dengan show ip interface brief

```

(s): Speeding
(d): Dampening Suppressed
(E): E-Trunk down
The number of interface that is UP in Physical is 23
The number of interface that is DOWN in Physical is 0
The number of interface that is UP in Protocol is 16
The number of interface that is DOWN in Protocol is 7
Interface          IP Address/Mask  Physical  Protocol VPN
Ethernet1/0/0       10.10.10.2/30    up        up      l3vpn
Ethernet1/0/0.4094  128.1.138.137/16 up        up      l3vpn
Ethernet1/0/1       unassigned       up        down    --
Ethernet1/0/1.4094  128.1.138.137/16 up        up      l3vpn
Ethernet1/0/2       192.168.30.1/24 up        up      l3vpn
Ethernet1/0/2.4094  128.1.138.137/16 up        up      l3vpn
Ethernet1/0/3       10.10.20.1/30    up        up      --
Ethernet1/0/3.4094  128.1.138.137/16 up        up      l3vpn
Ethernet1/0/4       10.10.11.2/30    up        up      --
Ethernet1/0/4.4094  128.1.138.137/16 up        up      l3vpn
... More ...

```

**Gambar 37:** Verifikasi interface router Huawei dengan display ip interface brief



```

[admin@MikroTik] > ip address print
Flags: X - disabled, I - invalid, D - dynamic
# ADDRESS NETWORK INTERFACE
0 :: TO-CISCO
10.10.30.2/30 10.10.30.0 ether2
1 10.10.20.2/30 10.10.20.0 ether1
[admin@MikroTik] >
[admin@MikroTik] >
[admin@MikroTik] >
[admin@MikroTik] >
[admin@MikroTik] >
[admin@MikroTik] >

```

**Gambar 38:** Verifikasi interface MikroTik dengan ip address print

## 9. Ping antar router

```

Router#ping 10.10.10.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/3 ms
Router#ping 10.10.11.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.11.2, timeout is 2 seconds:
!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/3 ms
Router#ping 10.10.30.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.30.2, timeout is 2 seconds:
!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/2 ms
Router#

```

**Gambar 39:** Ping dari router Cisco ke link Huawei dan MikroTik

```

[~HUAWEI]ping 10.10.10.1
PING 10.10.10.1: 56 data bytes, press CTRL C to break
Reply from 10.10.10.1: bytes=56 Sequence=1 ttl=255 time=4 ms
Reply from 10.10.10.1: bytes=56 Sequence=2 ttl=255 time=2 ms
Reply from 10.10.10.1: bytes=56 Sequence=3 ttl=255 time=2 ms
Reply from 10.10.10.1: bytes=56 Sequence=4 ttl=255 time=2 ms
Reply from 10.10.10.1: bytes=56 Sequence=5 ttl=255 time=2 ms

--- 10.10.10.1 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 2/2/4 ms

[~HUAWEI]ping 10.10.11.1
PING 10.10.11.1: 56 data bytes, press CTRL C to break
Reply from 10.10.11.1: bytes=56 Sequence=1 ttl=255 time=1 ms
Reply from 10.10.11.1: bytes=56 Sequence=2 ttl=255 time=1 ms
Reply from 10.10.11.1: bytes=56 Sequence=3 ttl=255 time=1 ms
Reply from 10.10.11.1: bytes=56 Sequence=4 ttl=255 time=1 ms

--- 10.10.11.1 ping statistics ---
4 packet(s) transmitted
4 packet(s) received
0.00% packet loss
round-trip min/avg/max = 1/1/2 ms

[~HUAWEI]

```

**Gambar 40:** Ping dari router Huawei ke link Cisco

```

[~HUAWEI]ping 10.10.20.2
PING 10.10.20.2: 56 data bytes, press CTRL C to break
Reply from 10.10.20.2: bytes=56 Sequence=1 ttl=64 time=1 ms
Reply from 10.10.20.2: bytes=56 Sequence=2 ttl=64 time=1 ms
Reply from 10.10.20.2: bytes=56 Sequence=3 ttl=64 time=1 ms
Reply from 10.10.20.2: bytes=56 Sequence=4 ttl=64 time=1 ms
Reply from 10.10.20.2: bytes=56 Sequence=5 ttl=64 time=2 ms

--- 10.10.20.2 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 1/1/2 ms

[~HUAWEI]

```

**Gambar 41:** Ping dari router Huawei ke link MikroTik

```

[admin@MikroTik] > ping 10.10.20.1
SEQ HOST SIZE TTL TIME STATUS
0 10.10.20.1 56 255 3ms
1 10.10.20.1 56 255 2ms
2 10.10.20.1 56 255 0ms
sent=3 received=3 packet-loss=0% min-rtt=0ms avg-rtt=1ms max-rtt=3ms

[admin@MikroTik] > ping 10.10.30.1
SEQ HOST SIZE TTL TIME STATUS
0 10.10.30.1 56 255 1ms
1 10.10.30.1 56 255 0ms
sent=2 received=2 packet-loss=0% min-rtt=0ms avg-rtt=0ms max-rtt=1ms

[admin@MikroTik] >

```

**Gambar 42:** Ping dari MikroTik ke link Huawei dan Cisco

## 1.4 Praktikum 4: Konfigurasi OSPF Multivendor dan OSPF Cost

1. Mengkonfigurasi OSPF pada Cisco Router (router-id, network statement area 0, dan pengaturan OSPF cost pada interface link ke Huawei dan MikroTik)

```

Router(config)#router ospf 1
Router(config-router)#router-id 1.1.1.1
Router(config-router)#network 10.10.10.0 0.0.0.3 area 0
Router(config-router)#network 10.10.11.0 0.0.0.3 area 0
Router(config-router)#network 10.10.30.0 0.0.0.3 area 0
Router(config-router)#network 192.168.10.0 0.0.0.255 area 0
Router(config-router)#network 192.168.20.0 0.0.0.255 area 0
Router(config-router)#network 192.168.30.0 0.0.0.255 area 0
Router(config-router)#network 192.168.40.0 0.0.0.255 area 0
Router(config-router)#exit
Router(config)#

```

**Gambar 43:** Konfigurasi OSPF pada router Cisco (router-id dan network area 0)

```

Router(config)#interface gi0/3
Router(config-if)#description LINK1-T0-HUAWEI
Router(config-if)#ip ospf cost 10
Router(config-if)#exit
Router(config)#interface gi0/0
Router(config-if)#description LINK2-T0-HUAWEI
Router(config-if)#ip ospf cost 10
Router(config-if)#exit
Router(config)#interface gi0/1
Router(config-if)#description LINK-T0-MIKROTIK
Router(config-if)#ip ospf cost 100
Router(config-if)#exit

```

**Gambar 44:** Pengaturan OSPF cost pada interface router Cisco

- Memverifikasi OSPF pada Cisco Router (show ip ospf neighbor, show ip route ospf, show ip protocols)

```

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.10.10.0 0.0.0.3 area 0
    10.10.11.0 0.0.0.3 area 0
    10.10.30.0 0.0.0.3 area 0
    192.168.10.0 0.0.0.255 area 0
    192.168.20.0 0.0.0.255 area 0
    192.168.30.0 0.0.0.255 area 0
    192.168.40.0 0.0.0.255 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
  Distance: (default is 110)

```

**Gambar 45:** Verifikasi OSPF pada Cisco dengan show ip protocols

- Mengkonfigurasi OSPF pada Huawei Router (router-id, network statement area 0.0.0.0, dan pengaturan OSPF cost pada interface link ke Cisco dan MikroTik)

```

[*HUAWEI]ospf 1 router-id 2.2.2.2
[*HUAWEI-ospf-1]area 0.0.0.0
[*HUAWEI-ospf-1-area-0.0.0.0]network 10.10.10.0 0.0.0.3
[*HUAWEI-ospf-1-area-0.0.0.0]network 10.10.11.0 0.0.0.3
[*HUAWEI-ospf-1-area-0.0.0.0]network 10.10.20.0 0.0.0.3
[*HUAWEI-ospf-1-area-0.0.0.0]network 192.168.50.0 0.0.0.255
[*HUAWEI-ospf-1-area-0.0.0.0]quit
[*HUAWEI-ospf-1]quit

```

**Gambar 46:** Konfigurasi OSPF pada router Huawei (router-id dan network area 0.0.0.0)

```

[*HUAWEI]interface Ethernet1/0/0
[*HUAWEI-Ethernet1/0/0]description LINK1-T0-CISCO
[*HUAWEI-Ethernet1/0/0]ospf cost 10
Error: Unrecognized command found at '^' position.
[*HUAWEI-Ethernet1/0/0]ospf cost 10
[*HUAWEI-Ethernet1/0/0]quit
[*HUAWEI]interface Ethernet1/0/4
[*HUAWEI-Ethernet1/0/4]description LINK2-T0-CISCO
[*HUAWEI-Ethernet1/0/4]ospf cost 10
[*HUAWEI-Ethernet1/0/4]quit
[*HUAWEI]interface Ethernet1/0/3
[*HUAWEI-Ethernet1/0/3]
Warning: Uncommitted configurations found, commit them before exiting? [Y(yes)/N(no)/C(cancel)]:C
[*HUAWEI-Ethernet1/0/3]description LINK-T0-MIKROTIK
[*HUAWEI-Ethernet1/0/3]ospf cost 100
[*HUAWEI-Ethernet1/0/3]quit
[*HUAWEI]commit

```

**Gambar 47:** Pengaturan OSPF cost pada interface router Huawei

- Memverifikasi OSPF pada Huawei Router (display ospf peer, display ip routing-table protocol ospf, display ospf routing)

```

[~HUAWEI]display ospf peer
(M) Indicates MADJ neighbor

      OSPF Process 1 with Router ID 2.2.2.2
      Neighbors

Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1      Address: 10.10.10.1
State: Full           Mode:Nbr is Slave      Priority: 1
DR: 10.10.10.1        BDR: 10.10.10.2      MTU: 1500
Dead timer due in 30 sec
Retrans timer interval: 5
Neighbor is up for 00h00m31s
Neighbor Up Time : 2026-06-05 03:47:04
Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
Router ID: 1.1.1.1      Address: 10.10.11.1
State: Full           Mode:Nbr is Slave      Priority: 1
DR: 10.10.11.1        BDR: 10.10.11.2      MTU: 1500
Dead timer due in 34 sec
Retrans timer interval: 5
Neighbor is up for 00h00m31s
Neighbor Up Time : 2026-06-05 03:47:04
Authentication Sequence: [ 0 ]

[~HUAWEI]

```

**Gambar 48:** Verifikasi OSPF peer pada Huawei dengan display ospf peer

```

[~HUAWEI]display ip routing-table protocol ospf
Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route
-----
public Routing Table : OSPF
Destinations : 9          Routes : 14

OSPF routing table status : <Active>
Destinations : 5          Routes : 10

Destination/Mask  Proto  Pre  Cost  Flags NextHop  Interface
-----
10.10.30.0/30    OSPF   10   110   D  10.10.11.1    Ethernet1/0/4
192.168.10.0/24  OSPF   10   110   D  10.10.10.1    Ethernet1/0/0
192.168.10.0/24  OSPF   10   11    D  10.10.11.1    Ethernet1/0/4
192.168.20.0/24  OSPF   10   11    D  10.10.10.1    Ethernet1/0/0
192.168.20.0/24  OSPF   10   11    D  10.10.11.1    Ethernet1/0/4
192.168.30.0/24  OSPF   10   11    D  10.10.10.1    Ethernet1/0/0
192.168.30.0/24  OSPF   10   11    D  10.10.11.1    Ethernet1/0/4
192.168.40.0/24  OSPF   10   11    D  10.10.10.1    Ethernet1/0/0
192.168.40.0/24  OSPF   10   11    D  10.10.11.1    Ethernet1/0/4

OSPF routing table status : <Inactive>
Destinations : 4          Routes : 4

Destination/Mask  Proto  Pre  Cost  Flags NextHop  Interface
-----
10.10.10.0/30    OSPF   10   10     10.10.10.2    Ethernet1/0/0
10.10.11.0/30    OSPF   10   10     10.10.11.2    Ethernet1/0/4

```

**Gambar 49:** Verifikasi routing table OSPF pada Huawei (display ip routing-table protocol ospf)

```

public Routing Table : OSPF
Destinations : 9          Routes : 14

OSPF routing table status : <Active>
Destinations : 5          Routes : 10

Destination/Mask  Proto  Pre  Cost  Flags NextHop  Interface
-----
10.10.30.0/30    OSPF   10   110   D  10.10.11.1    Ethernet1/0/4
192.168.10.0/24  OSPF   10   110   D  10.10.10.1    Ethernet1/0/0
192.168.10.0/24  OSPF   10   11    D  10.10.11.1    Ethernet1/0/4
192.168.20.0/24  OSPF   10   11    D  10.10.10.1    Ethernet1/0/0
192.168.20.0/24  OSPF   10   11    D  10.10.11.1    Ethernet1/0/4
192.168.30.0/24  OSPF   10   11    D  10.10.10.1    Ethernet1/0/0
192.168.30.0/24  OSPF   10   11    D  10.10.11.1    Ethernet1/0/4
192.168.40.0/24  OSPF   10   11    D  10.10.10.1    Ethernet1/0/0
192.168.40.0/24  OSPF   10   11    D  10.10.11.1    Ethernet1/0/4

OSPF routing table status : <Inactive>
Destinations : 4          Routes : 4

Destination/Mask  Proto  Pre  Cost  Flags NextHop  Interface
-----
10.10.10.0/30    OSPF   10   10     10.10.10.2    Ethernet1/0/0
10.10.11.0/30    OSPF   10   10     10.10.11.2    Ethernet1/0/4
10.10.20.0/30    OSPF   10   100    10.10.20.1    Ethernet1/0/3
192.168.50.0/24  OSPF   10   1     192.168.50.1  Ethernet1/0/2

```

**Gambar 50:** Lanjutan routing table OSPF pada Huawei

```

Destination      Cost  Type  NextHop  AdvRouter  Area
-----
10.10.10.0/30    10    Direct 10.10.10.2 2.2.2.2    0.0.0.0
10.10.11.0/30    10    Direct 10.10.11.2 2.2.2.2    0.0.0.0
10.10.20.0/30    100   Direct 10.10.20.1 2.2.2.2    0.0.0.0
10.10.30.0/30    110   Stub   10.10.11.1 1.1.1.1    0.0.0.0
10.10.30.0/30    110   Stub   10.10.10.1 1.1.1.1    0.0.0.0
192.168.10.0/24  11    Stub   10.10.11.1 1.1.1.1    0.0.0.0
192.168.10.0/24  11    Stub   10.10.10.1 1.1.1.1    0.0.0.0
192.168.20.0/24  11    Stub   10.10.11.1 1.1.1.1    0.0.0.0
192.168.20.0/24  11    Stub   10.10.10.1 1.1.1.1    0.0.0.0
192.168.30.0/24  11    Stub   10.10.11.1 1.1.1.1    0.0.0.0
192.168.30.0/24  11    Stub   10.10.10.1 1.1.1.1    0.0.0.0
192.168.40.0/24  11    Stub   10.10.11.1 1.1.1.1    0.0.0.0
192.168.40.0/24  11    Stub   10.10.10.1 1.1.1.1    0.0.0.0
192.168.50.0/24  1     Direct 192.168.50.1 2.2.2.2    0.0.0.0

Total Nets: 9
Intra Area: 9 Inter Area: 0 ASE: 0 NSSA: 0

OSPF Process 65534 with Router ID 128.1.138.137
Routing Tables

Routing for Network
Destination      Cost  Type  NextHop  AdvRouter  Area
-----
128.1.138.137/32 0     Direct 128.1.138.137 128.1.138.137 0.0.0.0

Total Nets: 1
Intra Area: 1 Inter Area: 0 ASE: 0 NSSA: 0
[~HUAWEI]

```

**Gambar 51:** Verifikasi OSPF routing pada Huawei dengan display ospf routing

5. Mengkonfigurasi OSPF pada MikroTik (router-id, network OSPF, dan pengaturan cost interface sebagai jalur backup)

(Konfigurasi OSPF MikroTik dilakukan melalui terminal sebelum verifikasi pada Gambar 52)

6. Memverifikasi OSPF pada MikroTik (/routing ospf neighbor print dan /ip route print where ospf)

```
[admin@MikroTik] > /routing ospf neighbor print
0 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 prio
rity=1
dr-address=10.10.30.1 backup-dr-address=0.0.0.0 state="Full" state-changes=
5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=9s
1 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 prio
rity=1
dr-address=10.10.20.1 backup-dr-address=10.10.20.2 state="Full" state-chang
es=5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=23s
[admin@MikroTik] >
```

**Gambar 52:** Verifikasi OSPF neighbor pada MikroTik dengan routing ospf neighbor print

```
[admin@MikroTik] > /ip route print where ospf
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS      PREF-SRC  GATEWAY      DISTANCE
0 ADo 10.10.10.0/30      10.10.30.1    110
1 ADo 10.10.11.0/30       10.10.20.1    110
2 ADo 192.168.10.0/24      10.10.30.1    110
3 ADo 192.168.20.0/24      10.10.30.1    110
4 ADo 192.168.30.0/24      10.10.30.1    110
5 ADo 192.168.40.0/24      10.10.30.1    110
6 ADo 192.168.50.0/24      10.10.20.1    110
[admin@MikroTik] >
```

**Gambar 53:** Verifikasi routing table OSPF pada MikroTik (ip route print where ospf)

7. Memverifikasi OSPF neighbor pada Cisco Router (show ip ospf neighbor; pastikan Huawei muncul dua kali dan MikroTik satu kali)

```
Router#show ip ospf neighbor
Neighbor ID    Pri  State           Dead Time   Address        Interface
3.3.3.3        1    FULL/BDR        00:00:34    10.10.30.2     GigabitEthernet0/1
2.2.2.2        1    FULL/BDR        00:00:33    10.10.11.2     GigabitEthernet0/0
2.2.2.2        1    FULL/BDR        00:00:33    10.10.10.2     GigabitEthernet0/3
Router#
```

**Gambar 54:** Verifikasi OSPF neighbor pada Cisco dengan show ip ospf neighbor

8. Memverifikasi OSPF peer pada Huawei Router (display ospf peer; pastikan peer Cisco muncul dua kali dan MikroTik satu kali)

```
(M) Indicates MADJ neighbor

OSPF Process 1 with Router ID 2.2.2.2
Neighbors
Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1      Address: 10.10.10.1
State: Full            Mode:Nbr is Slave      Priority: 1
DR: 10.10.10.1         BDR: 10.10.10.2        MTU: 1500
Dead timer due in 35 sec
Retrans timer interval: 5
Neighbor is up for 00h08m08s
Neighbor Up Time : 2026-06-05 03:47:04
Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.20.1 (Eth1/0/3)'s neighbors
Router ID: 3.3.3.3      Address: 10.10.20.2
State: Full            Mode:Nbr is Master     Priority: 1
DR: 10.10.20.1         BDR: 10.10.20.2        MTU: 1500
Dead timer due in 32 sec
Retrans timer interval: 5
Neighbor is up for 00h01m51s
```

**Gambar 55:** Verifikasi OSPF peer pada Huawei (Cisco dan MikroTik)

```

Area 0.0.0.0 interface 10.10.20.1 (Eth1/0/3)'s neighbors
Router ID: 3.3.3.3      Address: 10.10.20.2
State: Full            Mode:Nbr is Master    Priority: 1
DR: 10.10.20.1         BDR: 10.10.20.2       MTU: 1500
Dead timer due in 32 sec
Retrans timer interval: 5
Neighbor is up for 00h01m51s
Neighbor Up Time : 2026-06-05 03:53:21
Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
Router ID: 1.1.1.1      Address: 10.10.11.1
State: Full            Mode:Nbr is Slave    Priority: 1
DR: 10.10.11.1         BDR: 10.10.11.2       MTU: 1500
Dead timer due in 33 sec
Retrans timer interval: 5
Neighbor is up for 00h08m08s
Neighbor Up Time : 2026-06-05 03:47:04
Authentication Sequence: [ 0 ]

[~HUAWEI]
[~HUAWEI]

```

**Gambar 56:** Lanjutan verifikasi OSPF peer pada Huawei

9. Memverifikasi OSPF neighbor pada MikroTik (/routing ospf neighbor print; pastikan Cisco dan Huawei muncul)

```

[admin@Mikrotik] > /routing ospf neighbor print
0 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1
dr-address=10.10.30.1 backup-dr-address=10.10.30.2 state="Full" state-changes=5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=2m26s

1 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1
dr-address=10.10.20.1 backup-dr-address=10.10.20.2 state="Full" state-changes=5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=2m40s
[admin@Mikrotik] >

```

**Gambar 57:** Verifikasi OSPF neighbor pada MikroTik (Cisco dan Huawei)

10. Mengecek routing table pada Cisco Router menuju LAN Huawei (show ip route 192.168.50.0)

```

Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
Known via "ospf 1", distance 110, metric 11, type intra area
Last update from 10.10.11.2 on GigabitEthernet0/0, 00:06:10 ago
Routing Descriptor Blocks:
  10.10.11.2, from 2.2.2.2, 00:06:10 ago, via GigabitEthernet0/0
    Route metric is 11, traffic share count is 1
  * 10.10.10.2, from 2.2.2.2, 00:06:10 ago, via GigabitEthernet0/3
    Route metric is 11, traffic share count is 1
Router#

```

**Gambar 58:** Routing table Cisco menuju LAN Huawei (show ip route 192.168.50.0)

11. Mengecek seluruh route OSPF pada Cisco Router (show ip route ospf)

```

Router#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
O    10.10.20.0/30 [110/110] via 10.10.11.2, 00:06:17, GigabitEthernet0/0
    [110/110] via 10.10.10.2, 00:06:17, GigabitEthernet0/3
O    192.168.50.0/24 [110/11] via 10.10.11.2, 00:06:17, GigabitEthernet0/0
    [110/11] via 10.10.10.2, 00:06:17, GigabitEthernet0/3
Router#

```

**Gambar 59:** Routing table OSPF pada Cisco dengan show ip route ospf

12. Mengecek seluruh route OSPF pada Huawei Router (display ip routing-table protocol ospf)

```

OSPF routing table status : <Active>
Destinations : 5      Routes : 10

```

| Destination/Mask | Proto | Pre | Cost | Flags | NextHop    | Interface     |
|------------------|-------|-----|------|-------|------------|---------------|
| 10.10.30.0/30    | OSPF  | 10  | 110  | D     | 10.10.11.1 | Ethernet1/0/4 |
| 10.10.11.0/30    | OSPF  | 10  | 110  | D     | 10.10.10.1 | Ethernet1/0/0 |
| 192.168.10.0/24  | OSPF  | 10  | 11   | D     | 10.10.11.1 | Ethernet1/0/4 |
| 192.168.20.0/24  | OSPF  | 10  | 11   | D     | 10.10.10.1 | Ethernet1/0/0 |
| 192.168.30.0/24  | OSPF  | 10  | 11   | D     | 10.10.11.1 | Ethernet1/0/4 |
| 192.168.40.0/24  | OSPF  | 10  | 11   | D     | 10.10.10.1 | Ethernet1/0/0 |
| 192.168.50.0/24  | OSPF  | 10  | 11   | D     | 10.10.11.1 | Ethernet1/0/4 |
| 192.168.60.0/24  | OSPF  | 10  | 11   | D     | 10.10.10.1 | Ethernet1/0/0 |

```

OSPF routing table status : <Inactive>
Destinations : 4      Routes : 4

```

| Destination/Mask | Proto | Pre | Cost | Flags | NextHop    | Interface     |
|------------------|-------|-----|------|-------|------------|---------------|
| 10.10.10.0/30    | OSPF  | 10  | 10   |       | 10.10.10.2 | Ethernet1/0/0 |
| 10.10.11.0/30    | OSPF  | 10  | 10   |       | 10.10.11.2 | Ethernet1/0/4 |
| 10.10.20.0/30    | OSPF  | 10  | 100  |       | 10.10.20.1 | Ethernet1/0/3 |
| 192.168.50.0/24  | OSPF  | 10  | 1    |       | 10.10.20.1 | Ethernet1/0/3 |

**Gambar 60:** Routing table OSPF pada Huawei dengan display ip routing-table protocol ospf

| Destination/Mask | Proto | Pre | Cost | Flags | NextHop      | Interface     |
|------------------|-------|-----|------|-------|--------------|---------------|
| 10.10.10.0/30    | OSPF  | 10  | 10   |       | 10.10.10.2   | Ethernet1/0/0 |
| 10.10.11.0/30    | OSPF  | 10  | 10   |       | 10.10.11.2   | Ethernet1/0/4 |
| 10.10.20.0/30    | OSPF  | 10  | 100  |       | 10.10.20.1   | Ethernet1/0/3 |
| 192.168.50.0/24  | OSPF  | 10  | 1    |       | 192.168.50.1 | Ethernet1/0/2 |

**Gambar 61:** Lanjutan routing table OSPF pada Huawei

13. Mengecek seluruh route OSPF pada MikroTik (/ip route print where ospf; pastikan route menuju Cisco dan Huawei tersedia)

```

[admin@MikroTik] > ip route print where ospf
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, F - fib, B - bgp, O - ospf, M - mme,
B - blackhole, U - unreachable, P - prohibit

```

| # | DST-ADDRESS         | PREF-SRC | GATEWAY    | DISTANCE |
|---|---------------------|----------|------------|----------|
| 0 | Ado 10.10.10.0/30   |          | 10.10.30.1 | 110      |
| 1 | Ado 10.10.11.0/30   |          | 10.10.20.1 | 110      |
| 2 | Ado 192.168.10.0/24 |          | 10.10.30.1 | 110      |
| 3 | Ado 192.168.20.0/24 |          | 10.10.30.1 | 110      |
| 4 | Ado 192.168.30.0/24 |          | 10.10.30.1 | 110      |
| 5 | Ado 192.168.40.0/24 |          | 10.10.30.1 | 110      |
| 6 | Ado 192.168.50.0/24 |          | 10.10.20.1 | 110      |

**Gambar 62:** Routing table OSPF pada MikroTik (ip route print where ospf)

14. Verifikasi koneksi end-to-end: ping dari PC VLAN 10, 20, 30, dan 40 menuju LAN Huawei (192.168.50.10)

```

VPCS> ping 192.168.50.10

```

|                                                             |
|-------------------------------------------------------------|
| 84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=6.362 ms |
| 84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=2.623 ms |
| 84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=2.667 ms |
| 84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=3.965 ms |
| 84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=4.101 ms |

```

VPCS>

```

**Gambar 63:** Ping dari PC VLAN 10 ke LAN Huawei (192.168.50.10)

```

VPCS> ping 192.168.50.10

```

|                                                             |
|-------------------------------------------------------------|
| 84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=3.165 ms |
| 84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=3.912 ms |
| 84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=3.586 ms |
| 84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=4.995 ms |
| 84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=5.210 ms |

```

VPCS>
VPCS>

```

**Gambar 64:** Ping dari PC VLAN 20 ke LAN Huawei (192.168.50.10)

```
VPCS> ping 192.168.50.10
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=3.374 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=2.811 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=4.430 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=2.585 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=3.448 ms
```

**Gambar 65:** Ping dari PC VLAN 30 ke LAN Huawei (192.168.50.10)

```
VPCS> ping 192.168.50.10
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=3.591 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=5.208 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=3.803 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=4.044 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=2.442 ms
```

**Gambar 66:** Ping dari PC VLAN 40 ke LAN Huawei (192.168.50.10)

15. Verifikasi koneksi end-to-end: ping dari PC LAN Huawei menuju IP VLAN Cisco (192.168.10.11, 192.168.20.11, 192.168.30.10, 192.168.40.10)

```
VPCS> ping 192.168.50.1
84 bytes from 192.168.50.1 icmp_seq=1 ttl=255 time=7.575 ms
84 bytes from 192.168.50.1 icmp_seq=2 ttl=255 time=2.284 ms
84 bytes from 192.168.50.1 icmp_seq=3 ttl=255 time=2.423 ms
^C
VPCS> ping 192.168.10.11
84 bytes from 192.168.10.11 icmp_seq=1 ttl=62 time=3.340 ms
84 bytes from 192.168.10.11 icmp_seq=2 ttl=62 time=4.027 ms
84 bytes from 192.168.10.11 icmp_seq=3 ttl=62 time=3.950 ms
84 bytes from 192.168.10.11 icmp_seq=4 ttl=62 time=2.928 ms
^C
VPCS> ping 192.168.20.11
84 bytes from 192.168.20.11 icmp_seq=1 ttl=62 time=3.628 ms
84 bytes from 192.168.20.11 icmp_seq=2 ttl=62 time=3.822 ms
84 bytes from 192.168.20.11 icmp_seq=3 ttl=62 time=3.400 ms
84 bytes from 192.168.20.11 icmp_seq=4 ttl=62 time=4.134 ms
^C
VPCS> ping 192.168.30.10
84 bytes from 192.168.30.10 icmp_seq=1 ttl=62 time=4.126 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=62 time=2.618 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=62 time=4.254 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=62 time=3.323 ms
84 bytes from 192.168.30.10 icmp_seq=5 ttl=62 time=3.156 ms
^C
```

**Gambar 67:** Ping dari PC LAN Huawei ke gateway dan IP VLAN 10, 20, 30 Cisco

## 1.5 Praktikum 5: Pengujian Failover OSPF

1. Mengaktifkan kembali kedua link Cisco-Huawei (interface gi0/3 dan gi0/0 no shutdown).

```
VPCS> ping 192.168.40.10
84 bytes from 192.168.40.10 icmp_seq=1 ttl=62 time=3.816 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=62 time=4.136 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=62 time=3.909 ms
84 bytes from 192.168.40.10 icmp_seq=4 ttl=62 time=3.406 ms
84 bytes from 192.168.40.10 icmp_seq=5 ttl=62 time=4.513 ms
VPCS> 
```

**Gambar 68:** Ping dari PC LAN Huawei ke IP VLAN 40 Cisco (192.168.40.10)

## 2 Analisis Hasil Percobaan

Praktikum 1 berfokus pada segmentasi jaringan layer 2 menggunakan VLAN pada Cisco Switch. Konfigurasi VLAN 10 (Finance), 20 (HR), 30 (IT), dan 40 (Marketing) berhasil dibuat dan berstatus aktif, dibuktikan melalui `show vlan brief`. Setiap port menuju end device dikonfigurasi sebagai access port sesuai VLAN-nya (untagged), sementara interface menuju Cisco Router dikonfigurasi sebagai trunk dot1Q yang mengizinkan VLAN 10, 20, 30, dan 40 sekaligus, sesuai hasil `show interfaces`

trunk. Dengan konfigurasi ini, lalu lintas antar-VLAN tetap terisolasi meski berbagi satu jalur fisik trunk.

Praktikum 2 mengimplementasikan komunikasi antar-VLAN pada layer 3 melalui Router-on-a-Stick (ROAS). Interface fisik Gi0/2 pada Cisco Router dipecah menjadi subinterface Gi0/2.10, Gi0/2.20, Gi0/2.30, dan Gi0/2.40, masing-masing dengan encapsulation dot1q dan IP gateway sesuai VLAN. Router juga berperan sebagai DHCP server untuk VLAN 10 dan 20, sedangkan VLAN 30 dan 40 menggunakan IP statis. Hasil `show ip interface brief` menunjukkan seluruh subinterface aktif (up/up), dan pengujian dari VPCS membuktikan client VLAN 10 dan 20 berhasil memperoleh IP secara DHCP, seluruh VLAN dapat menjangkau gateway-nya, serta ping antar-VLAN dari VLAN 10 ke VLAN 20, 30, dan 40 berhasil tanpa packet loss.

Praktikum 3 membangun koneksi point-to-point multivendor antara Cisco, Huawei, dan MikroTik menggunakan subnet /30 pada setiap link. LAN tambahan diaktifkan pada Huawei (Ethernet1/0/2, 192.168.50.0/24). Perbedaan sintaks konfigurasi antar-vendor cukup terlihat, seperti penggunaan `system-view` dan `commit` pada Huawei dibandingkan mode konfigurasi langsung pada Cisco dan pendekatan berbasis `command path` pada MikroTik. Verifikasi `show/display ip interface brief` dan `ip address print` menunjukkan seluruh interface penghubung berstatus up, dan ping antar-router (Cisco-Huawei, Huawei-MikroTik, Cisco-MikroTik) berhasil. Pada tahap ini, konektivitas masih terbatas pada perangkat yang terhubung langsung karena routing dinamis belum diaktifkan.

Praktikum 4 mengintegrasikan seluruh jaringan menggunakan OSPF single-area (Area 0). Setiap router mengiklankan network yang terhubung langsung, dengan link Cisco-Huawei diberi cost 10 sebagai jalur utama dan link melalui MikroTik diberi cost 100 sebagai jalur backup. Verifikasi `show ip ospf neighbor` dan `display ospf peer` menunjukkan neighbor antar-router mencapai status FULL — Huawei muncul dua kali pada Cisco (dua link utama) dan MikroTik satu kali. Routing table pada Cisco menampilkan dua next-hop berbeda menuju 192.168.50.0/24 dengan cost sama, menandakan Equal-Cost Multi-Path (ECMP) berjalan. Pengujian end-to-end membuktikan seluruh PC pada VLAN Cisco dapat berkomunikasi dengan PC pada LAN Huawei, dan sebaliknya.

Praktikum 5 menguji mekanisme failover OSPF dengan menonaktifkan link utama Cisco-Huawei secara bertahap. Pada kondisi normal (dua link aktif), trace dari PC VLAN ke 192.168.50.10 melewati jalur langsung Cisco-Huawei sesuai ECMP. Setelah satu link dimatikan, trace tetap melewati jalur langsung melalui link yang tersisa, menunjukkan OSPF berhasil melakukan reconvergence tanpa kehilangan konektivitas. Ketika kedua link utama dimatikan, jalur routing berpindah ke rute backup melalui MikroTik (cost 100), dengan trace menunjukkan tambahan hop PC VLAN → Cisco → MikroTik → Huawei → LAN Huawei. Hasil ini membuktikan bahwa pengaturan cost OSPF efektif mengarahkan trafik ke jalur cadangan hanya ketika jalur utama benar-benar tidak tersedia, sesuai prinsip routing berbasis cost terendah.

### 3 Hasil Tugas Modul

Tugas modul tambahan dikerjakan dan dokumentasinya dapat dilihat pada file README repository kelompok di GitHub.



## 4 Kesimpulan

Rangkaian Praktikum 1 hingga 5 berhasil mendemonstrasikan pembangunan jaringan multivendor (Cisco, Huawei, MikroTik) secara bertahap dari layer 2 hingga layer 3. Segmentasi VLAN dan trunking 802.1Q pada layer 2 berhasil memisahkan empat domain broadcast (Finance, HR, IT, Marketing) sekaligus tetap dapat dilewatkan melalui satu jalur trunk. Pada layer 3, ROAS memungkinkan satu interface fisik melayani routing antar seluruh VLAN, sementara DHCP server menyederhanakan pengalamatan IP untuk sebagian client.

Koneksi point-to-point antar-vendor menggunakan subnet /30 berhasil dibangun sebagai dasar interkoneksi, meskipun pada tahap ini konektivitas masih terbatas pada jaringan yang terhubung langsung. Penerapan OSPF Area 0 kemudian menyatukan seluruh segmen menjadi satu jaringan yang saling terhubung secara otomatis, dengan pengaturan cost yang berhasil membentuk jalur utama (Cisco-Huawei, cost 10) dan jalur cadangan (via MikroTik, cost 100). Fitur ECMP turut berjalan pada dua link berbiaya sama, mendukung load balancing.

Pengujian failover pada Praktikum 5 membuktikan bahwa OSPF mampu melakukan reconvergence secara otomatis ketika jalur utama terputus, dengan trafik berpindah ke jalur backup tanpa intervensi manual pada tabel routing. Secara keseluruhan, seluruh tujuan praktikum tercapai: jaringan multivendor yang dibangun terbukti fungsional, efisien dalam penggunaan jalur, dan andal terhadap kegagalan link.